Mykola Shatokhin

S1. Model parameters and diagnostics

The model consists of agents nested in groups, six cultural domains and five prerequisite levels. Reproduction samples parents according to repertoire benefit and cognitive cost. Learners observe local or external models, copy prerequisite chains with level-dependent error, recombine the best domain levels from multiple models and innovate. Teaching, adaptive model number and infrastructure are continuous functions of observed repertoires, not binary switches. Table S1 lists the reference configuration; Figures S1 and S2 report global parameter sensitivity and outcome-definition robustness.

Table S1. Reference model configuration and outcome diagnostics.

Quantity	Reference value
Groups x agents	16 x 20
Domains x levels	6 x 5
Generations	420
Years per generation	25
Baseline external-model probability	0.002
Primary individual diagnostic	S >= 16 and >=5 developed domains
Persistent population diagnostic	>50% agents for 45/50 final generations; mean S >18

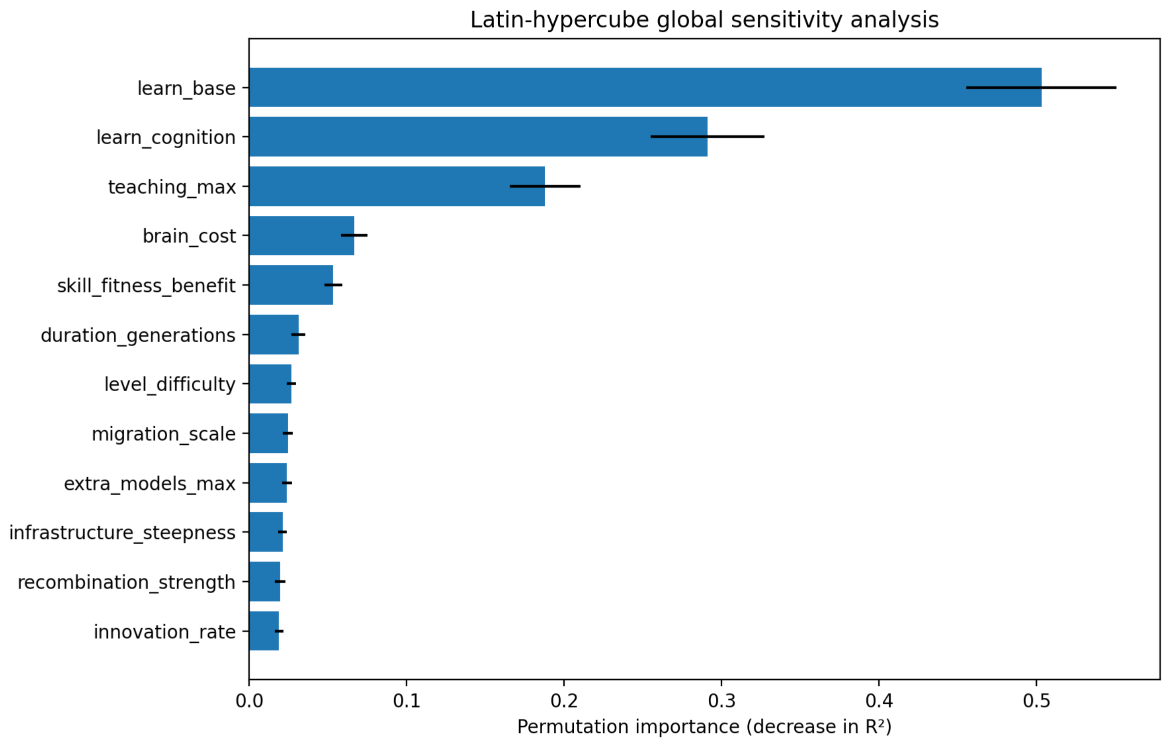

Figure 1: Bar plot of permutation importance and associations for fourteen model parameters.

Figure S1. Global Latin-hypercube sensitivity analysis.

Figure S2. Ignition boundaries under permissive, primary and strict diagnostics.

S2. Expanded ablation results

Table S2 reports duration-specific boundaries for the full model and each ablation in the expanded grid. This grid uses four contact values and 50 replicates per cell, whereas the primary dense phase experiment uses nine

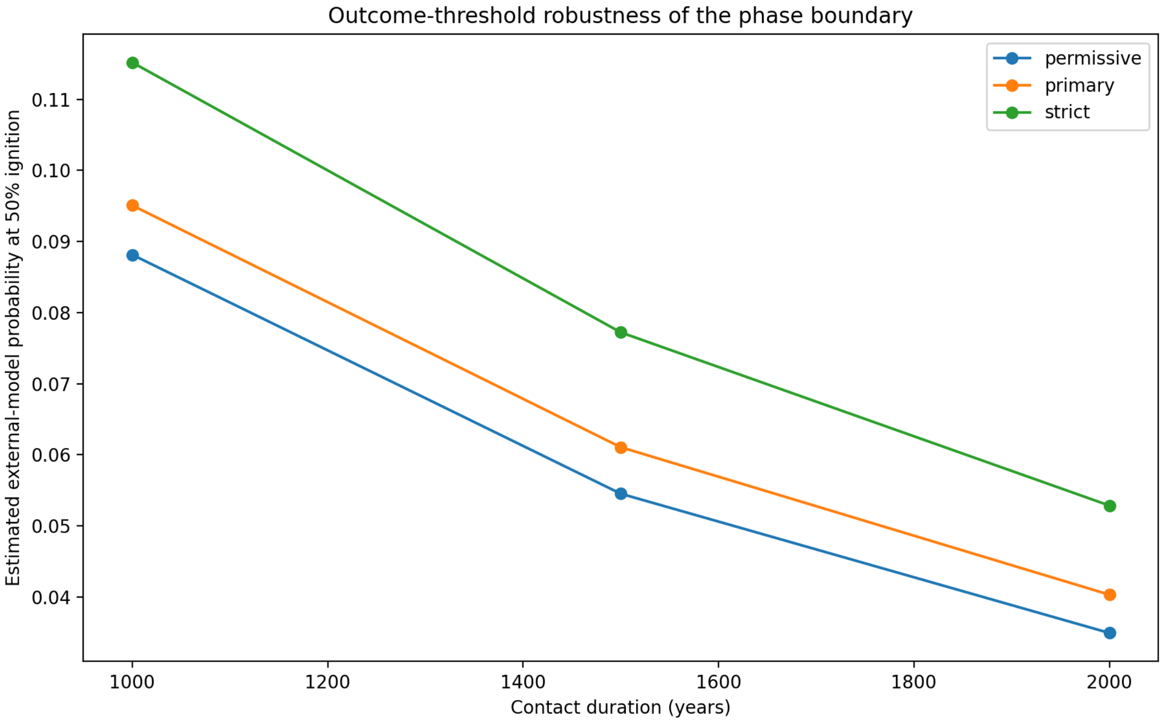

Figure 2: Curves compare the estimated phase boundary under three outcome definitions.

contact values and 100 replicates per cell. The full-model values in Table S2 are therefore separate robustness estimates, not replacements for the dense phase thresholds reported in the main text.

Table S2. Expanded ablation boundaries across contact duration.

Model	Duration (years)	Estimated p50	Maximum observed P
Full model	1000	0.107	0.72
Full model	1500	0.064	0.88
Full model	2000	0.041	0.94
No cultural selection on cognition	1000	1.124	0.02
No cultural selection on cognition	1500	0.217	0.18
No cultural selection on cognition	2000	0.193	0.26
No infrastructure feedback	1000	not reached	0.00
No infrastructure feedback	1500	not reached	0.00
No infrastructure feedback	2000	not reached	0.00
No recombination	1000	not reached	0.00
No recombination	1500	not reached	0.00
No recombination	2000	not reached	0.00
No teaching	1000	not reached	0.00
No teaching	1500	not reached	0.00
No teaching	2000	not reached	0.00

Zero ignition in the tested grid establishes component dependence only for $p_{\text{ext}} \leq 0.16$, contact duration ≤ 2000 years and the remaining reference parameters of this model architecture. It does not establish universal biological necessity and does not preclude ignition under untested extremes or alternative mechanisms.

S3. Archaeological robustness

Table S3 summarizes date and cluster uncertainty. Figures S3 and S4 show the permutation null and threshold sensitivity; Table S4 gives the fixed-boundary results.

Table S3. Archaeological change-point estimates under date and cluster uncertainty.

Analysis	Median change point	95% interval
Midpoint optimum	525 ka	-
Dates sampled in intervals	500 ka	325-575 ka
Source-cluster bootstrap	525 ka	275-575 ka
Site-cluster bootstrap	525 ka	275-575 ka

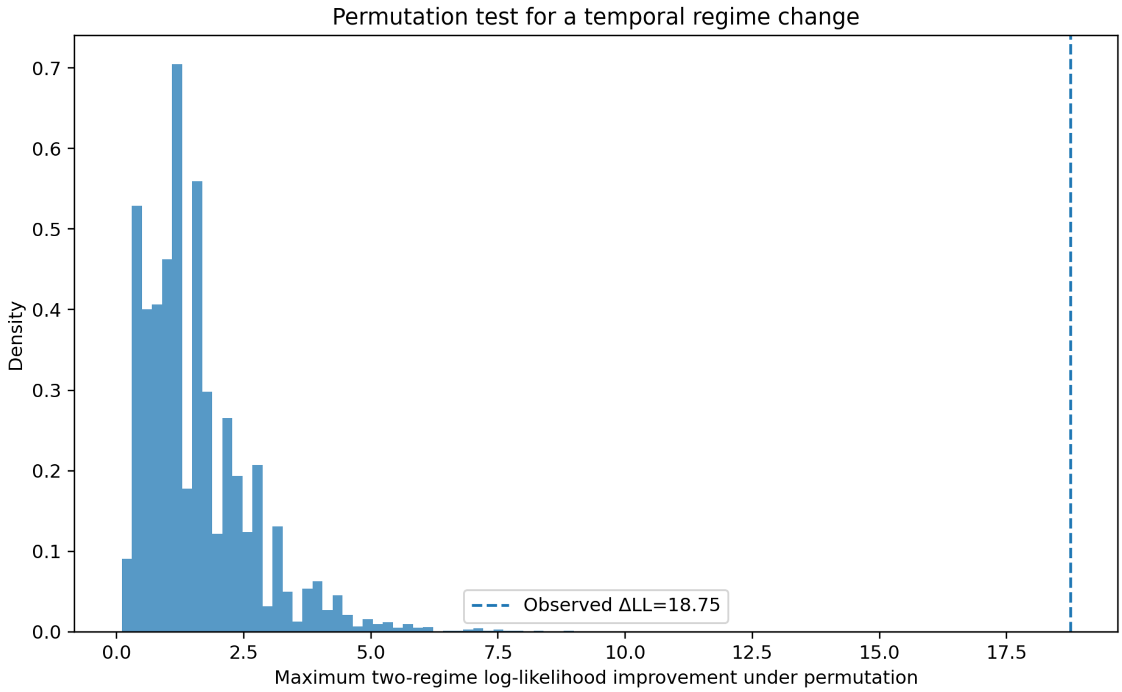

Figure 3: Histogram of permutation delta log-likelihood values with the observed value marked beyond the distribution.

Figure S3. Temporal permutation test for the archaeological change point.

Figure S4. Sensitivity to procedural baselines and fixed temporal boundaries.

Table S4. Sensitivity to procedural baseline and temporal boundary.

Baseline	Boundary (ka)	Older rate	Younger rate	Fisher p
5	500	0.471	0.979	7.67e-06
5	525	0.438	0.979	3.68e-06
5	600	0.357	0.980	6.67e-07
5	700	0.357	0.980	6.67e-07
6	500	0.235	0.957	1.63e-08
6	525	0.188	0.958	4.00e-09
6	600	0.143	0.940	1.13e-08
6	700	0.143	0.940	1.13e-08
7	500	0.118	0.894	1.08e-08
7	525	0.062	0.896	1.41e-09
7	600	0.000	0.880	8.10e-10
7	700	0.000	0.880	8.10e-10

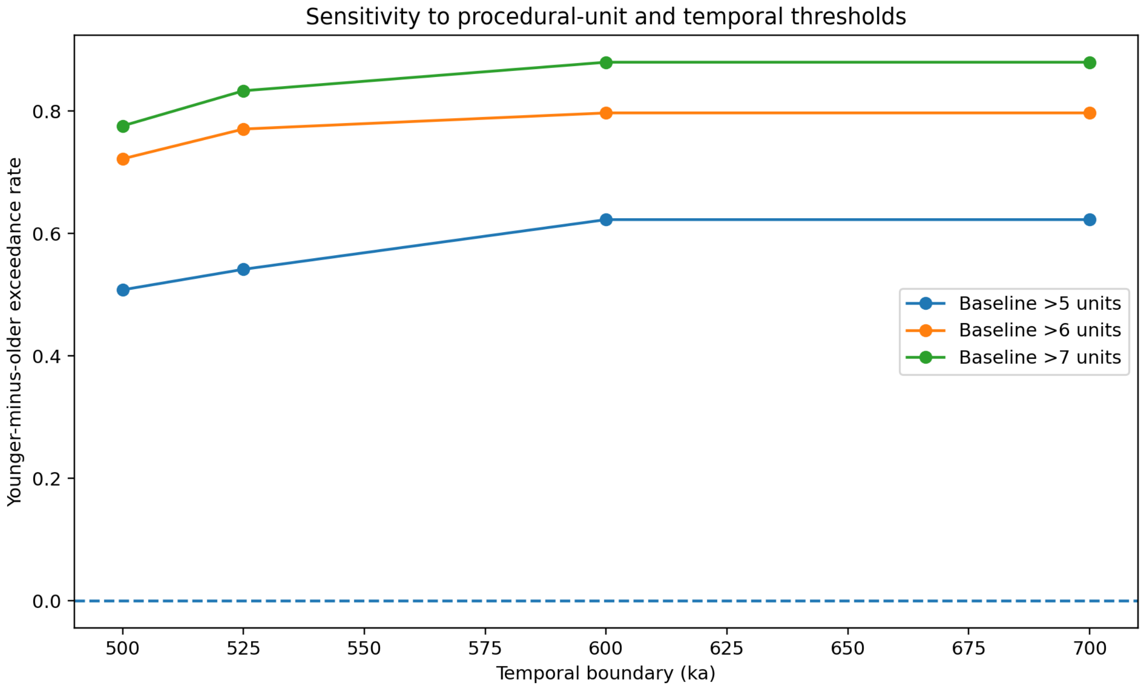

Figure 4: Plot compares older and younger exceedance rates under different baselines and temporal boundaries.

S4. Network-realization uncertainty

For stochastic topologies, twenty graphs were independently generated within every contact stratum, and each graph was paired with eight agent seeds. Because the graphs differ across contact strata, the bootstrap resamples graphs independently within each stratum before calculating pooled p50. It does not construct artificial graph-specific trajectories across contact levels. Figure S5 shows the pooled curves, and Table S5 reports network-metric associations.

Figure S5. Pooled ignition curves for nine network topologies.

Table S5. Associations between network metrics and contact-adjusted ignition.

Network metric	Spearman rho with contact-adjusted ignition	p
density	0.571	2.78e-53
clustering	0.315	2.59e-15
mean_shortest_path	-0.515	7.30e-42
degree_cv	-0.407	2.54e-25
modularity	-0.467	9.08e-34

S5. Population-architecture robustness

Table S6 lists the exploratory scaling configurations. Figures S6-S8 separate fixed-total, group-count and group-size effects.

Table S6. Population-architecture robustness ensembles and estimated thresholds.

Design	Groups	Group size	N	p50	Status
fixed_group_20	4	20	80		>0.12
fixed_group_20	8	20	160	0.067	estimated
fixed_group_20	16	20	320	0.040	estimated
fixed_group_20	32	20	640	0.050	estimated
fixed_group_20	64	20	1280	0.037	estimated
fixed_groups_16	16	10	160	0.093	estimated

Design	Groups	Group size	N	p50	Status
fixed_groups_16	16	20	320	0.043	estimated
fixed_groups_16	16	40	640		<0.03
fixed_groups_16	16	80	1280		<0.03
fixed_total_320	4	80	320		>0.12
fixed_total_320	8	40	320	0.063	estimated
fixed_total_320	16	20	320	0.040	estimated
fixed_total_320	32	10	320	0.070	estimated

Figure S6. Architecture at fixed total population.

Figure S7. Robustness to group count with 20 agents per group.

Figure S8. Robustness to local group size with 16 groups.

S6. Cosmological derivation and scenarios

Assume successful cultural-ignition events across eligible complex biospheres form a Poisson process. The mean eligibility exposure $\bar{E} = \mathbb{E}[E \mid \text{complex life}]$ is not estimated from the archaeological or agent-based data. It is an explicit scenario parameter defined as the mean eligible duration conditional on a rocky HZ planet already having reached complex animal-grade life. The separate factor f_{complex} accounts for the fraction of HZ planets that reach that state, preventing double counting. The fiducial calculation uses $\bar{E} = 4$ Gyr, and the repository also evaluates 1 and 10 Gyr. All bounds scale exactly as $1/\bar{E}$. The local ABM probability multiplies, rather than replaces, the opportunity rate. The firstness probability is conditional on these assumptions and does not incorporate SETI survey selection, settlement, detectability or civilisation survival. Table S7 gives the fiducial filter constraints, Table S8 shows exposure sensitivity, and Figure S9 shows how contact-dependent P_{ignite} changes the conditional probability of no earlier successful event.

Table S7. Conditional Great-Filter bounds across Galactic planet-count scenarios.

NHZ	Criterion	Maximum $f_{\text{complex}} \times r_{\text{opp}}$ (Gyr^{-1})
1e+08	50% first	3.466e-09
1e+08	95% first	2.565e-10
1e+08	Expected prior <1	5.000e-09
1e+09	50% first	3.466e-10
1e+09	95% first	2.565e-11
1e+09	Expected prior <1	5.000e-10
1e+10	50% first	3.466e-11
1e+10	95% first	2.565e-12
1e+10	Expected prior <1	5.000e-11

Table S8. Sensitivity of the 50% firstness bound to mean eligible exposure for $N_{\text{HZ}} = 10^{10}$ and $P_{\text{ignite}} = 0.5$.

$\bar{E}$ (Gyr)	Maximum $f_{\text{complex}} r_{\text{opp}}$ (Gyr^{-1})	Relative to fiducial
1	1.386e-10	4.0×
4	3.466e-11	1.0×
10	1.386e-11	0.4×

Figure S9. Conditional firstness constraints as contact changes Pignite.

S7. Reproducibility architecture

The package separates exact reproduction of published summaries from computationally expensive stochastic regeneration. The preprocessing script `prepare_archaeological_data.py` rebuilds the 64-row analysis table from the bundled Windows-1252 source CSV by applying the documented archaeological and single-chain filters, summing the 33 procedural indicators, and calculating age midpoints. Frozen ensemble CSV files are the

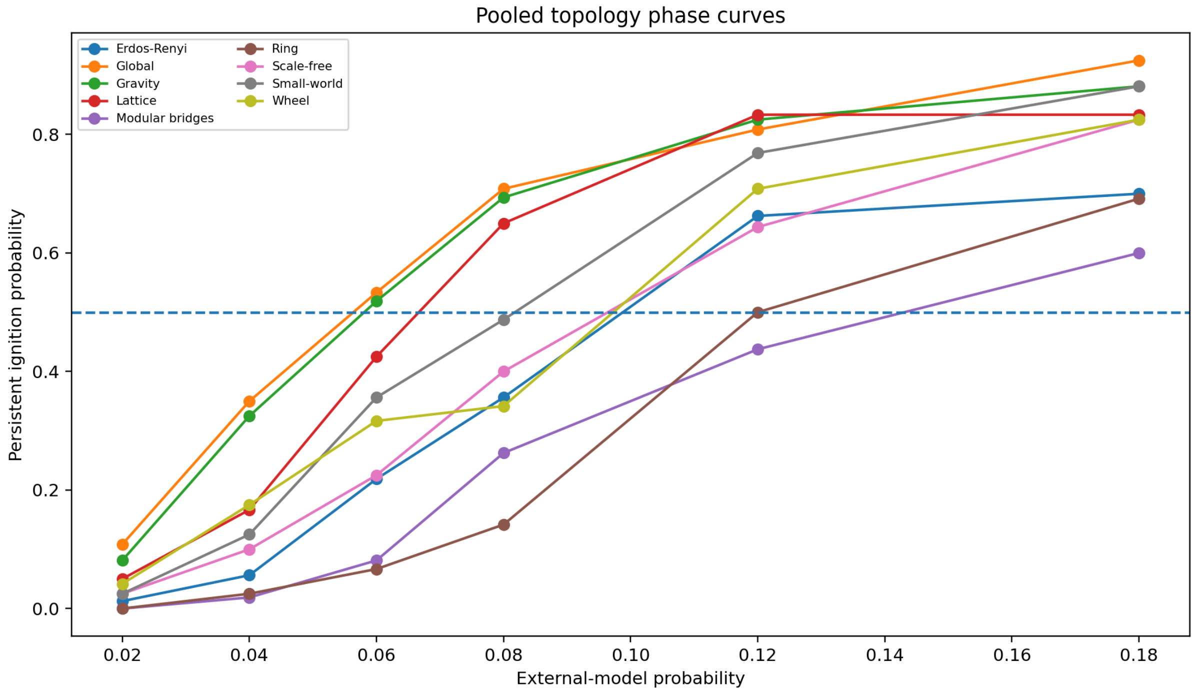

Figure 5: Nine curves show persistent ignition probability as external model probability increases.

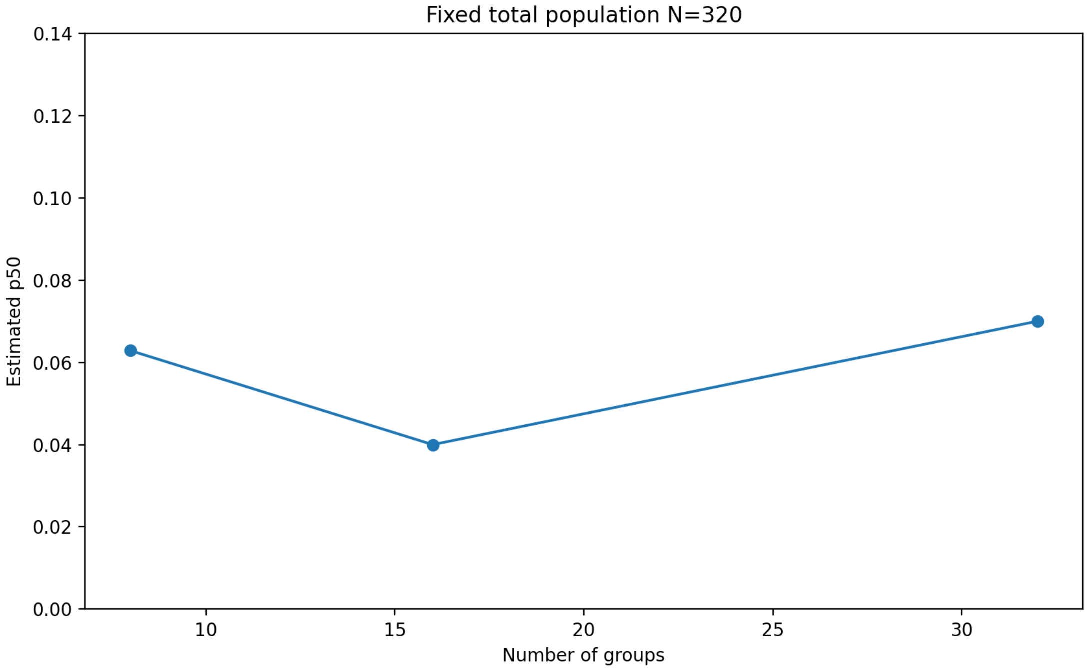

Figure 6: Line plot of estimated threshold for group partitions with total population 320.

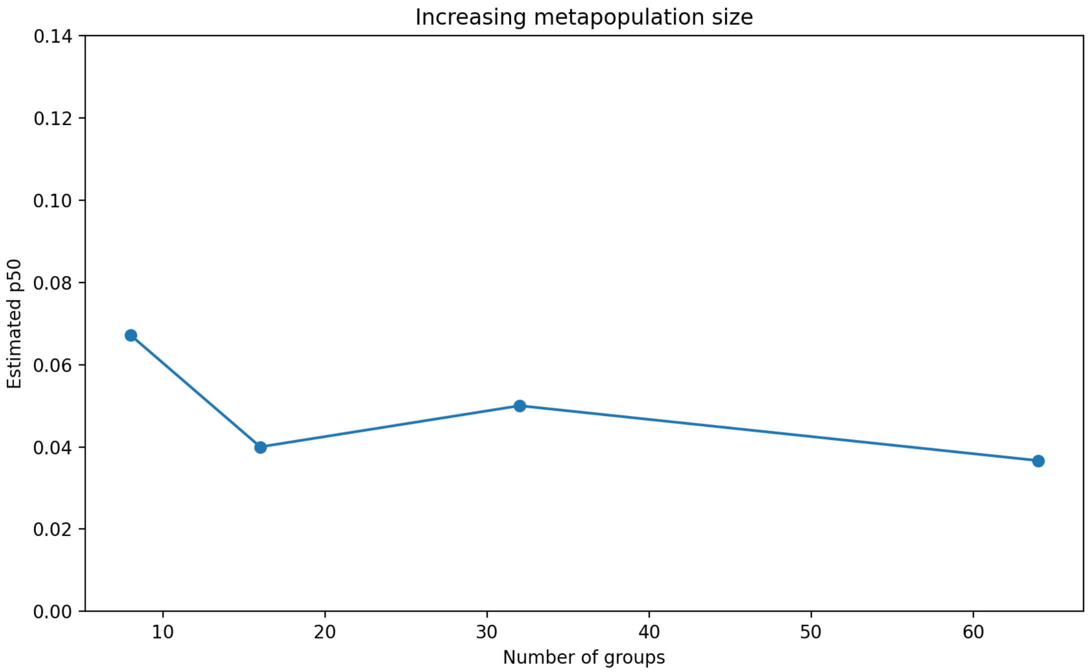

Figure 7: Line plot of estimated threshold versus number of groups.

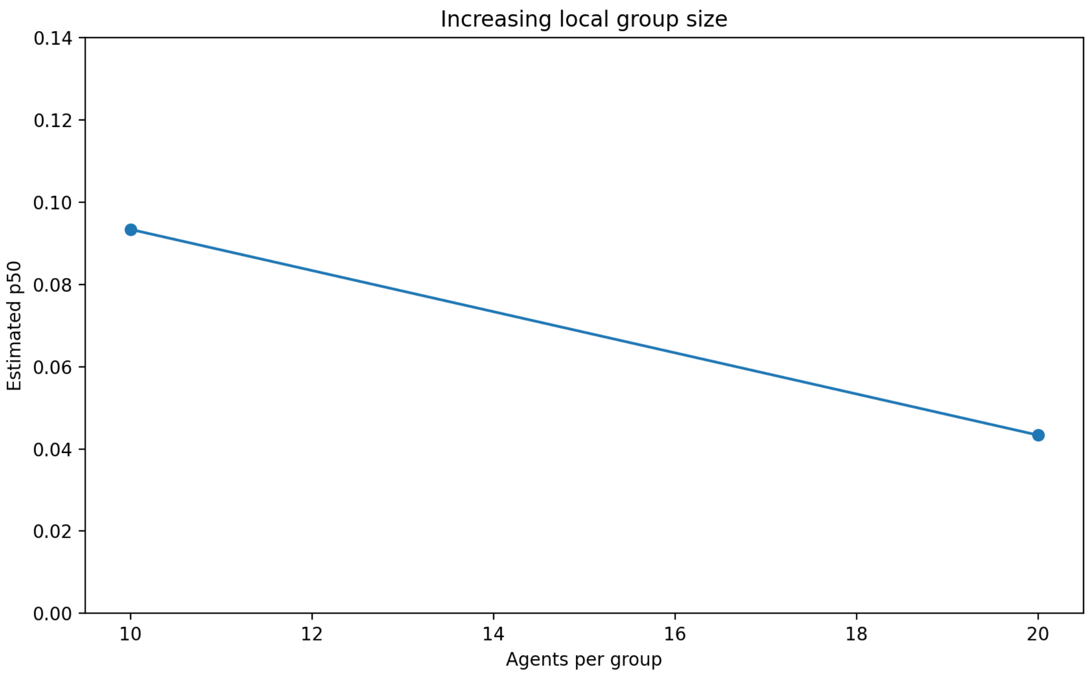

Figure 8: Line plot of estimated threshold versus agents per group.

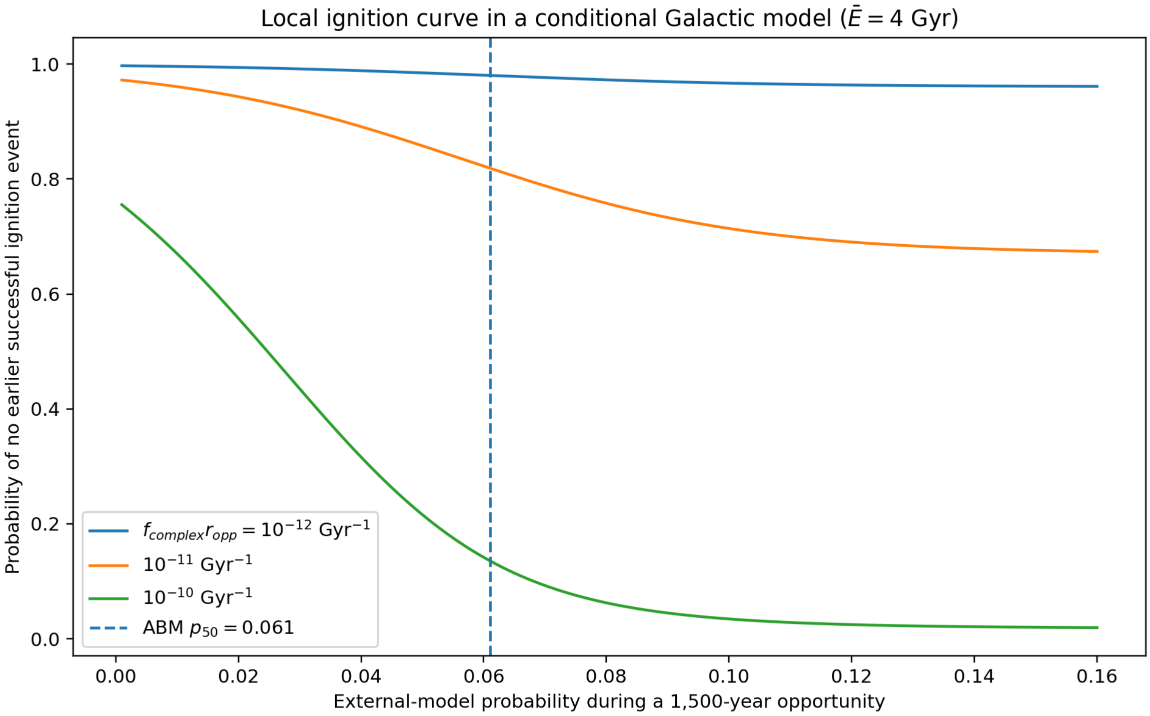

Figure 9: Conditional probability of no earlier successful ignition event as external cultural contact changes the local ABM ignition probability. Curves show three assumed products $f_{\text{complex}} r_{\text{opp}}$ for $N_{\text{HZ}} = 10^{10}$ and $\bar{E} = 4$ Gyr.

publication records. The script reproduce_published_analyses.py rebuilds the derived archaeological table and then regenerates archaeological, ablation, topology, sensitivity, scaling and cosmological summaries and figures. Stable-seed runners are provided for fresh simulations. The scaling publication runner uses exactly eight replicates per point, matching the frozen scaling ensemble; the previous mismatched 20/15 runner has been removed from the publication package.

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